Homework #4 Byoungchan Lee
ECON5345, Spring 2026
Due: March 8, 2026, 10:00 PM

Reminder: Empirical and simulation exercises must be submitted with "stand-alone" code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

1. Download the monthly trade-weighted US dollar exchange rate (TWEXBPA at FRED). Apply Phillips-Perron, augmented Dickey-Fuller, and KPSS tests to the logged exchange rate (pptest, adftest, kpsstest in Matlab). Discuss your results. Repeat for real GDP.

2. Consider an n-dimensional $VAR(p)$ model. We write the model using the notation on slide 12.

$$y_t = \Pi x_t + e_t.$$

The first row of the above equation, when stacked over t, can be written as follows:

$$y_1 = \begin{pmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{1T} \end{pmatrix} = \begin{pmatrix} x_1' \\ x_2' \\ \vdots \\ x_T' \end{pmatrix} \beta_1 + \begin{pmatrix} e_{11} \\ e_{12} \\ \vdots \\ e_{1T} \end{pmatrix} = X\beta_1 + e_1,$$

where β_1 is the first row of Π (with a transpose). Now, by stacking the above equations, we have

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} X & 0 & \cdots & 0 \\ 0 & X & \cdots & \vdots \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & X \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = (I_n \otimes X)\beta + e,$$

where $\otimes$ denotes the Kronecker product.

a) Show that $var(e) = \Omega \otimes I_T$, where $\Omega = var(v_t)$ and $v_t = (e_{1t}, e_{2t}, \dots, e_{nt})'$. (Hint: v_t is white noise. So, $cov(v_t, v_\tau) = 0$ for any $t \neq \tau$.)

b) Show that $\hat{\beta}_{GLS} = \left((I_n \otimes X)'(\Omega \otimes I_T)^{-1}(I_n \otimes X)\right)^{-1}((I_n \otimes X)'(\Omega \otimes I_T)^{-1}y)$ is equal to $\hat{\beta}_{OLS} = \left((I_n \otimes X)'(I_n \otimes X)\right)^{-1}((I_n \otimes X)'y)$. You may want to use the following properties of the Kronecker product: 1) $(A \otimes B)' = A' \otimes B'$, 2) $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$, and 3) $(A \otimes B)(C \otimes D) = (AC \otimes BD)$, when each operation is well-defined.

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3. Consider a three-variable VAR using quarterly observations: Δlog(real GDP), Δlog(CPI), FFR. To minimize the effects of the binding zero lower bound, you may use the sample up to 2008:q4.

a) Select the lag length using BIC, AIC, and likelihood ratio tests. Compare the differences.
b) Given the optimal lag length selected by AIC, estimate the model by least squares.
c) Using Cholesky decomposition and the ordering [Δlog(real GDP), Δlog(CPI), FFR],
   i. Compute and plot impulse responses of log(real GDP), log(CPI), FFR (horizon $h = 0,\ldots,20$) to a one-standard-deviation shock in FFR. (HINT: You can obtain the response of log(real GDP) by cumulating the responses of Δlog(real GDP)).
   ii. For each impulse response, construct, plot, and compare error bands (use 5th and 95th percentiles) using:
      – Parametric Monte-Carlo (HINT: You may want to use the half-vectorization routine, vech.m, and the duplication matrix, dupmat.m, uploaded on Canvas),
      – Bootstrap.
   iii. Does the price level increase or decrease in response to the contractionary monetary policy shock (i.e. after an increase in FFR)? How can you reconcile an increase (decrease) in the price level? You may want to consult Sims (EER, 1992).
   iv. Compute and plot the contribution of the FFR shocks to variation in log(real GDP), log(CPI), and FFR (the forecast variance decompositions) at horizons $h = 0,\ldots,20$.